

RAMCO INSTITUTE OF TECHNOLOGY
DEPARTMENT OF MECHANICAL ENGINEERING
ACADEMIC YEAR 2021-22
ME8811-PROJECT WORK

Sl.No.	Project Title	PO's Relevance	PSO's Relevance	Outcome
Solution to Industrial Problems based Projects				
1.	Energy Auditing in Nail Production Industry.	1,2,3,4,5,6,7,8,9,10,11,12	3,4	The students did their project in GowriHouse Metal Works, Rajapalayam. The project is appreciated and their suggestions will be implemented in the industry.
2.	Implementation of Quality Management System in Screw Manufacturing Industry.	1,2,3,4,5,6,7,8,9,10,11,12	1,2,4	Nalliraj Madhavan M and Thojesh Nandha M presented a paper on the title “A Review on Quality Control Tools in manufacturing Industries” in International Conference on Sustainable Materials, Manufacturing and Industrial Engineering (ICSMMIE) 2022 organised by Siddaganga Institute of Technology on 2nd July, 2022.
3.	Implementation of TQM to Improve Ergonomic Conditions in Howa Ring Frame.	1,2,3,4,5,6,7,8,9,10,11,12	1,2,4	- Rajasekar P and Rajajothi M presented a paper on the title “Implementation of TQM in Manufacturing Industry – A Critical Review” in International Conference on Sustainable Materials, Manufacturing and Industrial Engineering (ICSMMIE) 2022 organised by Siddaganga Institute of Technology on 2nd July, 2022. T. Ram Kumar, B. Subura Bala, P. Rajasekar and M. Rajajothi, IV Mech presented a paper titled as “Implementation of in TQM In Manufacturing Industry” in 20th Annual ISTE Tamilnadu Section Engineering Students Convention on Sustainable Development of India Through Innovations organised by Ramco Institute of Technology on 26th March 2022.

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4.	Performance Enhancement of Cone Winding Savio Polar Machine.	1,2,3,4,5,6,7,8,9,10,11,12	1,2,4	Ramkumar T and Subra Bala B presented a paper on the title "Implementation of TQM in Manufacturing Industry – A Critical Review" in International Conference on Sustainable Materials, Manufacturing and Industrial Engineering (ICSMMIE) 2022 organised by Siddaganga Institute of Technology on 2 nd July, 2022.
5.	Implementation of KANBAN System in Gowri House Metal Works LLP.	1,2,3,4,5,6,7,8,9,10,11,12	1,2,4	Sridharan S and Ragul G R presented a paper on the title "A review on implementation of Kanban system in various process industries" in International Conference On Smart Technologies and Applications (ICSTA 2022). Organised by Ramco Institute of Technology on 12.03.2022. The paper is also published in AIP Conference Proceedings.
6.	Implementation of e-TPM in Screw Manufacturing Industry.	1,2,3,4,5,6,7,8,9,10,11,12	1,2,4	• Ram Hari Prasad S and Prasanna Venkatesh H presented a paper on the title "A Review on Quality Control Tools in manufacturing Industries" in International Conference on Sustainable Materials, Manufacturing and Industrial Engineering (ICSMMIE) 2022 organised by Siddaganga Institute of Technology on 2nd July, 2022. • M. Nalliraj Madhavan, M. Thojesh Nandha, H. Prasanna venkatesh and S. Ram hari Prasad, presented a paper titled "Implementation of E-TPM in Medium Scale Industry" in 20th Annual ISTE Tamilnadu Section Engineering Students Convention on Sustainable Development of India through Innovations organised by Ramco Institute of Technology on 26th March 2022.
7.	Automatic Cone Packaging Machine.	1,2,3,4,5 ,8,9,10,11,12	1	The project is completed and the industry appreciated the students work.

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Projects in Identified Researches				
8.	Experimental Investigation of Drying Herbal Leaves using Solar Parabolic Trough Collector.	1,2,3,4,5,6,7,8,9,10,11,12	1,3	The project work is completed and will be published in an Annexure- I journal.
9.	Computational Simulation and Experimental Analysis of Fully Developed Flow Green House Solar Dryer at different Air Velocity.	1,2,3,4,5,6,7,8,9,10,11,12	3,4	The project work is completed. It is a part of faculty research work; journal writing is in progress and will be published soon in a SCI Journal.
10.	Erosion and Corrosion Behaviour of Plasma Sprayed WC+8YSZ Composite Coating on SS316.	1,2,3,4,5, 8,9,10,11,12	2	The Project work is completed and will be published in a reputed Journal.
11.	Optimization of Process Parameter of Plasma Sprayed Coating on SS316 using Cr ₃ C ₂ Feed Stock Powder for Corrosion Resistance.	1,2,3,4,5, 8,9,10,11,12	2	The Project work is completed and will be published in a reputed Journal.
12.	Erosion and Corrosion Wear Behaviour of Plasma Sprayed Composite Coating (Cr ₃ C ₂ +8YSZ) on SS316.	1,2,3,4,5, 8,9,10,11,12	2	The Project work is completed and will be published in a reputed Journal.
13.	Corrosion Resistance in Magnesium Alloy using EIS.	1,2,3,4,5, 8,9,10,11,12	2	The Project work is completed and will be published in a reputed Journal.
14.	Testing Analysis of ABS and Carbon reinforced ABS via FDM.	1,2,3,4,5, 8,9,10,11,12	2,4	The Project work is completed and will be published in a reputed Journal.
15.	Design and Analysis of Air Conditioning System for a Multi Storey Building for Different Climatic Condition.	1,2,3,4,5,6,7,8,9,10,11,12	1,3,4	The Project work is completed and will be published in a reputed Journal.
16.	Mixed Mode Fracture Behaviour on Aluminium Metal Matrix Composition.	1,2,3,4,5, 8,9,10,11,12	2	The Project work is completed and will be published in a reputed Journal.

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17.	Optimization of Processing Parameter of Plasma Sprayed Coating on SS316 using Tungsten Carbide with Chromium Carbide Feed stock Powder for Wear Resistance.	1,2,3,4,5, 8,9,10,11,12	2,4	The Project work is completed and will be published in a reputed Journal.
18.	Optimizing the Machining Parameters of Micro EDM for Food Grade Steel.	1,2,3,4,5 6,7, 8,9,10,11,12	2,4	The Project work is completed and will be published in a reputed Journal.
19.	Investigation Study of Cutting Parameters during Turning Hardened Steel with Carbide Inserts.	1,2,3,4,5, 8,9,10,11,12	2	The Project work is completed and will be published in a reputed Journal.
20.	Fracture Study on Magnesium Metal Matrix Composites.	1,2,3,4,5, 8,9,10,11,12	2	The Project work is completed and will be published in a reputed Journal.
21.	ECAP Testing for Magnesium based Composite.	1,2,3,4,5, 8,9,10,11,12	2	The Project work is completed and will be published in a reputed Journal.
22.	Enhancing the Wear Behaviour of Copper Composite by using Stir Casting Technique.	1,2,3,4,5,6,7,8,9,10,11,12	1,4	The Project work is completed and will be published in a reputed Journal.
23.	Mechanical Characterization and Finite Element Analysis of PETG Material via Additive Manufacturing (FDM) Process.	1,2,3,4,5,6,7,8,9,10,11,12	1,2,4	The Project work is completed and will be published in a reputed Journal.
24.	Mathematical Modelling of Electroless Ni-P Composite Coating using COMSOL.	1,2,3,4,5,6,7,8,9,10,11,12	1,4	The project is completed and the paper is under review in the Journal Surface Review Letters. It is submitted on 19.10.2022.
25.	Characterization of Additively Manufactured SS316 & 17-4PH.	1,2,3,4,5,6,7,8,9,10,11,12	1,2,4	The Project work is completed and will be published in a reputed Journal.
26.	Enhancing the Behaviour of Copper-Vanadium by using Stir Casting Technique.	1,2,3,4,5, 8,9,10,11,12	2	The Project work is completed and will be published in a reputed Journal.

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27.	Improving the Productivity of Solar Still by Enhancing Rate of Evaporation and Condensation.	1,2,3,4,5,6,7,8,9,10,11,12	3,4	Veereswaran N and Vishnu Sankar R presented a paper on the title "Progress in Improving the Productivity of Solarstill – A Review"" in International Conference On Smart Technologies and Applications (ICSTA 2022). Organised by Ramco Institute of Technology on 11.03.2022. The paper is also published in AIP Conference Proceedings.
28.	Enhancement in Evaporation Rate of Single Basin Single Slope Passive Solar Still.	1,2,3,4,5,6,7,8,9,10,11,12	3,4	Selvabalaji M and Palani Bharathi R presented a paper on the title "Energy and Exergy Analysis of Single Slope Passive Solar Still with Atomizer" in International Conference On Smart Technologies and Applications (ICSTA 2022). Organised by Ramco Institute of Technology on 11.03.2022. The paper is also published in AIP Conference Proceedings.
29.	Performance Study of Single Basin Solar Still with Evaporation Improvement Techniques.	1,2,3,4,5,6,7,8,9,10,11,12	3,4	The Project work is completed and will be published in a reputed Journal.
30.	Microstructure Study and Thermographic Analysis for the Crumb Rubber Composite.	1,2,3,4,5,6,7,8,9,10,11,12	1,2,4	Seenivasan K and Pragatheeswaran A Ponmuthumani J and Vishnu G presented a paper titled "Microstructure Study and Thermographic Analysis for the Crumb Rubber Composite." in International Conference on "Interdisciplinary Academic Research and Innovation" (IARI – 22) on 18.06.2022.
31.	Investigation of Mechanical Behaviour of Al Based Metal Matrix Composites.	1,2,3,4,5,6,7,8,9,10,11,12	1,4	The Project work is completed and will be published in a reputed Journal.
32.	Design and Analysis of Air Conditioning System for a Multi-Storey Building for different Climatic Condition.	1,2,3,4,5,6,7,8,9,10,11,12	3,4	The Project work is completed and will be published in a reputed Journal.

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33.	Coating on Cutting Tool Inserts under PVD Techniques.	1,2,3,4,5,6,7,8,9,10,11,12	1,2,4	The Project work is completed and will be published in a reputed Journal.
34.	Strengthening of Magnesium Composite by Adding Ceramic and Rare Earth Metal.	1,2,3,4,5,6,7,8,9,10,11,12	1,4	The Project work is completed and will be published in a reputed Journal.
35.	Numerical Simulation and Experimentation on Carbon Fibre Reinforced Polymer Under Buckling/Impact Behaviour of Military Aircraft's parts using Abacus/Python Script.	1,2,3,4,5,6,7,8,9,10,11,12	1,4	The Project work is completed and will be published in a reputed Journal.
36.	Analysis of Mechanical Properties of Recycled Crumb Rubber Sheet from Waste Tyre.	1,2,3,4,5,6,7,8,9,10,11,12	1,2,4	Rajasimha Reddy S, Ponmuthumani J and Vishnu G presented a paper titled "Analysis of Mechanical Properties of Recycled Crumb Rubber Sheet from Waste Tyre" in International Conference on "Interdisciplinary Academic Research and Innovation"(IARI – 22) on 18.06.2022.
Projects in Specific Application				
37.	Smart Solid Waste Segregating Bin Bot using a PLC.	1,2,3,4,5,6,7,8,9,10,11,12	1,2	The project work is completed and will be published in a journal.
38.	Food Waste Prediction in Hostel Campus using Deep Learning Models.	1,2,3,4,5,6,7,8,9,10,11,12	2	The project work is completed and the paper will be presented in a Conference.
39.	Manually Operated Sewer Waste Remover.	1,2,3,4,5,6,7,8,9,10,11,12	1,2	The project work is completed and patent drafting is in progress.
40.	Automatic Sanitizing Machine and Temperature Sensing.	1,2,3,4,5,6,7,8,9,10,11,12	1, 3	The project work is completed and patent drafting is in progress.
41.	IOT Based Smart Trash Bot Segregating Solid Waste.	1,2,3,4,5,6,7,8,9,10,11,12	1,2,4	The project work is completed and the paper will be presented in a Conference.
42.	Piezo Electric Energy Harvester.	1,2,3,4,5,6,7,8,9,10,11,12	1,2,3,4	The project work is completed and patent drafting is in progress.

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43.	Ground Nut Pod Separator.	1,2,3,4,5,6,7,8,9,10,11,12	1,2,4	This project was a part of IIT RuTAG Project.
44.	Characterization of Additively Manufactured Implant.	1,2,3,4,5,6,7,8,9,10,11,12	1,2,4	The project work is completed and will be published in a journal.
45.	Prediction of Solid Waste Generation in Metro City Using Deep Learning.	1,2,3,4,5,6,7,8,9,10,11,12	1,2,4	T. Samrath, M. Vigneshkumar presented a paper titled "Solid Waste Prediction in Metro City using Machine Learning Models" in Fourth International Conference on Advances in Industrial Engineering Applications (ICAIEA-2022) organised by the Department of Industrial Engineering, Anna University (CEG Campus) during 02-04 June, 2022.
46.	Bio Degradable Waste Segregator Machine.	1,2,3,4,5,6,7,8,9,10,11,12	1,2,4	The project work is completed and patent drafting is in progress.
47.	Non Bio Degradable Waste Segregator Machine.	1,2,3,4,5,6,7,8,9,10,11,12	1,2,4	The project work is completed and patent drafting is in progress.